

Stage 2: Progress Review

Diabetes Risk & Lifestyle Advisor
An AI-Based Decision-Support System

Course: Essentials of Artificial Intelligence

Due: Week 8 | Weight: 10%

Disclaimer: This system is a screening tool built for educational purposes only. It is NOT a substitute for professional medical diagnosis or treatment.

1. Updated Project Title and Problem Statement

Project Title

Diabetes Risk & Lifestyle Advisor - An AI-Based Decision-Support System

Problem Statement

Diabetes is a chronic metabolic disease affecting hundreds of millions worldwide, yet many cases go undiagnosed until complications arise. Early risk screening and personalized lifestyle guidance can significantly improve outcomes. However, existing screening tools typically rely on a single technique (e.g., a standalone questionnaire or a basic ML classifier), producing either a binary yes/no answer or a raw probability - neither of which is intuitive or actionable for patients.

Our system addresses this gap by combining three complementary AI techniques into a single integrated pipeline:

- A Neural Network that predicts diabetes risk probability from clinical health metrics.
- A Fuzzy Logic engine that translates the raw probability into a human-interpretable risk category (Low / Moderate / High / Severe), factoring in BMI and physical activity level.
- A Rule-Based Expert System that generates personalized, context-aware lifestyle and medical recommendations based on the risk category and the patient's specific health metrics.

The system is deployed as an interactive Streamlit web dashboard, making it accessible and easy to use for non-technical users.

2. Changes Made After Proposal Feedback

#	Feedback Received	Change Implemented
1	Need clearer integration between AI techniques	Redesigned as a sequential pipeline: NN -> Fuzzy -> Rules
2	Rule-based system felt too generic/static	Added metric-specific modifiers so recommendations are personalized
3	Fuzzy logic should consider more than just NN output	Expanded fuzzy inputs to 3 variables with 36 rule combinations
4	Dataset needs proper cleaning	Implemented zero-imputation and StandardScaler normalization
5	Include ethical considerations and disclaimers	Added medical disclaimer + documented dataset bias in sidebar/README
6	Need a proper test suite	Created test_pipeline.py with module tests + 10 full pipeline test profiles

3. Finalized AI Techniques

Technique 1: Neural Network (scikit-learn MLPClassifier)

- Purpose: Predict diabetes risk probability from 8 clinical health metrics
- Architecture: 2 hidden layers (16 neurons -> 8 neurons), ReLU activation, Adam optimizer

- Training: 500 max iterations, 80/20 stratified train-test split, random_state=42
- Preprocessing: StandardScaler normalization; zeros imputed with column medians
- Output: Continuous probability value (0.0-1.0) passed to the fuzzy engine

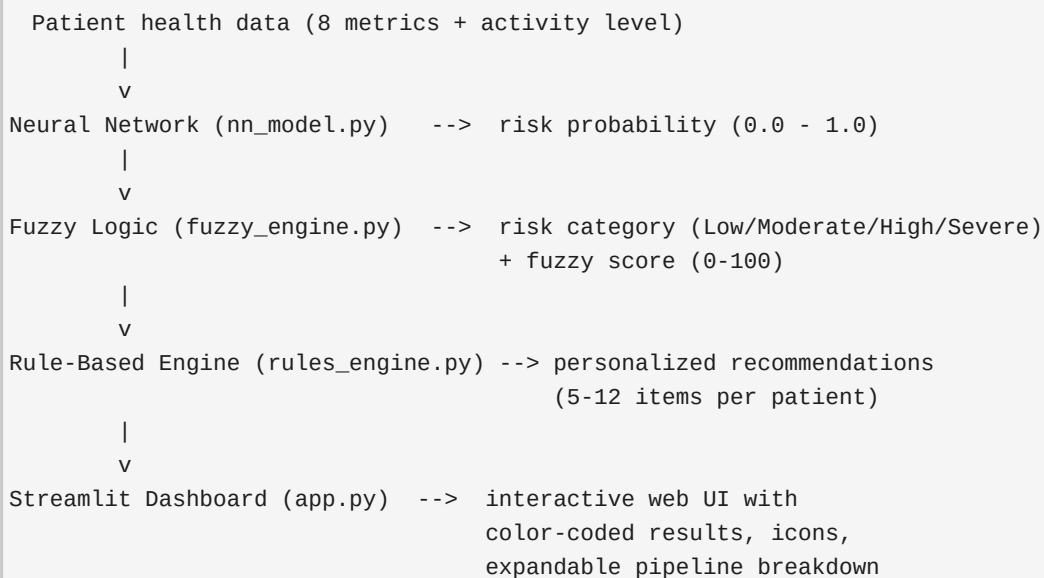
Technique 2: Fuzzy Logic (scikit-fuzzy)

- Purpose: Convert NN probability into a human-interpretable risk category
- Inputs (3 Antecedents): risk_probability, BMI, activity_level
- Output (1 Consequent): risk_category score (0-100) mapped to Low/Moderate/High/Severe
- Membership Functions: Trapezoidal and triangular (trapmf / trimf)
- Rules: 36 fuzzy rules covering all input combinations
- Defuzzification: Centroid method

Technique 3: Rule-Based Expert System (Plain Python)

- Purpose: Generate personalized, actionable lifestyle and medical recommendations
- 4 risk tiers with tier-specific base recommendations
- 6 metric-specific modifier blocks (glucose, BMI, BP, insulin, age, family history, activity)
- ~25-30 distinct recommendation strings, contextually selected
- Output: Ordered list beginning with a medical disclaimer

4. System Architecture / Workflow Diagram



Each stage feeds into the next - the fuzzy engine needs the NN's output, and the rule engine needs the fuzzy engine's category. The Streamlit dashboard orchestrates the pipeline and displays the results interactively.

5. Dataset Details

Property	Detail
Name	Pima Indians Diabetes Dataset
Source	NIDDK / UCI Machine Learning Repository
Records	768
Features	8 numeric predictors
Target	Binary outcome (0 = no diabetes, 1 = diabetes)
File	diabetes.csv (24 KB)

Feature Descriptions

Feature	Description	Range
Pregnancies	Number of times pregnant	0-17
Glucose	Plasma glucose concentration (2h OGTT, mg/dL)	0-199
BloodPressure	Diastolic blood pressure (mm Hg)	0-122
SkinThickness	Triceps skin fold thickness (mm)	0-99
Insulin	2-Hour serum insulin (uU/mL)	0-846
BMI	Body Mass Index (kg/m ²)	0-67.1
DiabetesPedigreeFunction	Diabetes pedigree function	0.078-2.42
Age	Age in years	21-81
Outcome	Class label	0, 1

Data Cleaning

- Columns where 0 is biologically impossible (Glucose, BloodPressure, SkinThickness, Insulin, BMI) have zeros replaced with NaN, then filled with the column median.

Known Limitations

- Dataset represents only adult women (≥ 21 years) of Pima Native American heritage from a 1960s study.
- Model may not generalize to men, other ethnicities, or younger populations.
- Class imbalance: ~65% negative, ~35% positive.

6. AI Technique Design Details

6.1 Neural Network Structure

Input Layer (8 neurons - one per health metric)
|
Hidden Layer 1 (16 neurons, ReLU activation)
|
Hidden Layer 2 (8 neurons, ReLU activation)
|
Output Layer (2 neurons, softmax - class probabilities)

Parameter	Value
Hidden layers	(16, 8)
Activation	ReLU
Optimizer	Adam
Max iterations	500
Random state	42
Train/test split	80/20 (stratified)
Scaler	StandardScaler

Model Evaluation Results

Metric	Score
Accuracy	0.7208 (72.08%)
Precision	0.5965 (59.65%)
Recall	0.6296 (62.96%)
F1 Score	0.6126 (61.26%)

Confusion Matrix

	Predicted Negative	Predicted Positive
Actual Negative	77 (TN)	23 (FP)
Actual Positive	20 (FN)	34 (TP)

6.2 Fuzzy Logic Variables and Rules

Antecedent Variables (Inputs):

Variable	Universe	Membership Functions
risk_probability	[0, 1.0] step 0.01	low (trapmf), medium (trimf), high (trapmf)

Variable	Universe	Membership Functions
bmi	[0, 60] step 1	underweight, normal, overweight, obese
activity_level	[0, 10] step 1	sedentary, moderate, active

Consequent Variable (Output):

Variable	Universe	Membership Functions
risk_category	[0, 100] step 1	low, moderate, high, severe

Sample Fuzzy Rules (36 total):

#	IF risk_prob	AND bmi	AND activity	THEN risk
1	high	obese	sedentary	severe
2	high	obese	active	high
3	high	normal	active	moderate
4	medium	overweight	moderate	moderate
5	medium	normal	active	low
6	low	obese	sedentary	moderate
7	low	normal	moderate	low

6.3 Rule-Based Engine Design

Tier-Based Rules:

Risk Category	# Base Rules	Example Recommendation
Low	4	Maintain your current healthy lifestyle.
Moderate	4	Aim to increase physical activity to 150 min/week.
High	5	Follow a structured low-glycemic index dietary plan.
Severe	5	Seek medical consultation immediately.

Metric-Specific Modifier Rules (6 blocks):

Metric	Condition	Action
Glucose	> 180 mg/dL	Seek medical attention promptly.
Glucose	> 140 mg/dL	Reduce refined carbohydrates.
BMI	> 35	Weight management critical - consult dietitian.
BMI	> 30	Structured weight management program.
BMI	< 18.5	Ensure adequate caloric intake.
Blood Pressure	> 100 mm Hg	Antihypertensive evaluation needed.
Blood Pressure	> 90 mm Hg	Reduce sodium, manage stress.
Insulin	> 200 uU/mL	Possible insulin resistance.
Age	> 60	Health check-ups every 6 months.

Metric	Condition	Action
Age	> 45	Ensure annual screening.
Family History	> 0.8	Genetic counseling may be beneficial.
Activity Level	< 3	Start with short daily walks.

7. Screenshots / Sample Outputs of Partial Implementation

7.1 Neural Network Training Output

```

Training model...
Model Evaluation:
Accuracy:  0.7208
Precision: 0.5965
Recall:    0.6296
F1 Score:  0.6126
Confusion Matrix:
[[77 23]
 [20 34]]

Testing predict_risk with sample patient data:
Predicted Diabetes Risk: 0.5070 (50.70%)

```

7.2 Individual Module Tests - All Passing

```

--- Neural Network Module ---
[PASS] predict_risk() returned 0.7073 (valid range)

--- Fuzzy Logic Module ---
[PASS] categorize_risk(0.1, 22.0, 8.0) -> Low (score: 11.67)
[PASS] categorize_risk(0.5, 30.0, 5.0) -> Moderate (score: 40.00)
[PASS] categorize_risk(0.9, 40.0, 1.0) -> Severe (score: 90.71)

--- Rule-Based Engine Module ---
[PASS] get_recommendations('Low', ...) -> 9 recommendations
[PASS] get_recommendations('Moderate', ...) -> 9 recommendations
[PASS] get_recommendations('High', ...) -> 10 recommendations
[PASS] get_recommendations('Severe', ...) -> 10 recommendations

[PASS] All individual module tests passed!

```

7.3 Full Pipeline Test - 10 Diverse Profiles

Profile	NN Prob	Fuzzy Score	Category	# Recs
Healthy Young Woman	0.0008	11.90	Low	5
Average Middle-Aged	0.1814	14.26	Low	4
Overweight Sedentary	0.8121	88.93	Severe	10
High-Risk Elderly	0.9749	90.71	Severe	12
Borderline Case A	0.4875	40.00	Moderate	4
Borderline Case B	0.7418	68.06	High	6
Very Active, Family History	0.0521	12.25	Low	6
Extreme High Risk	0.9397	90.71	Severe	12

Profile	NN Prob	Fuzzy Score	Category	# Recs
Young Underweight	0.0002	13.21	Low	5
High BP, Normal Glucose	0.1448	13.21	Low	6

7.4 Detailed Pipeline Output - Healthy Young Woman (Low Risk)

Input: Pregnancies=0, Glucose=85, BP=66, BMI=21.5, Age=22, Activity=8
 NN Risk Probability: 0.0008 (0.1%)
 Fuzzy Category: Low (score: 11.90)
 Recommendations (5):

1. Maintain your current healthy lifestyle.
2. Schedule an annual routine health checkup.
3. Stay adequately hydrated.
4. Continue eating a balanced diet.
5. Great job maintaining an active lifestyle!

7.5 Detailed Pipeline Output - High-Risk Elderly (Severe Risk)

Input: Pregnancies=8, Glucose=190, BP=95, BMI=39, Age=62, Activity=2
 NN Risk Probability: 0.9749 (97.5%)
 Fuzzy Category: Severe (score: 90.71)
 Recommendations (12):

1. Seek medical consultation immediately.
2. Daily blood glucose monitoring is mandatory.
3. A strict dietary plan prescribed by a dietitian is needed.
4. Medication or insulin therapy may be required.
5. Regular medical follow-ups every 2 weeks.
6. Glucose level significantly elevated.
7. BMI indicates severe obesity - weight management critical.
8. Blood pressure elevated - reduce sodium, manage stress.
9. Insulin levels elevated - possible insulin resistance.
10. Comprehensive health check-ups every 6 months.
11. Strong family history - genetic counseling beneficial.
12. Activity level low - start with short daily walks.

7.6 Streamlit Dashboard

The Streamlit web dashboard (app.py) is fully implemented and functional. Features include: a 3-column input form for all 8 health metrics plus activity level; an 'Analyze Risk' button triggering the full pipeline; color-coded risk category badges (green/yellow/red/dark-red); icon-tagged recommendations; an expandable JSON breakdown showing inputs/outputs of each pipeline stage; and a sidebar with dataset info, limitations, and tech stack.

8. Problems Faced During Development

#	Problem	Resolution
1	Biologically impossible zeros in dataset (Glucose=0, BMI=0)	Zero-to-NaN conversion + median imputation for 5 columns

#	Problem	Resolution
2	scikit-fuzzy crashes at universe boundaries	Input clamping: risk [0.01,0.99], BMI [1,59], activity [1,9]
3	Fuzzy rules not firing for some input combos	Try/except fallback using raw NN probability scaled to 0-100
4	Model not generalizing well initially	Tuned to (16,8) hidden layers + 500 iterations + StandardScaler
5	Recommendations felt too generic	Added 6 metric-specific modifier blocks for personalization
6	Import path issues (Streamlit vs direct)	sys.path.insert(0,...) in app.py and test_pipeline.py
7	Stale fuzzy simulation state between runs	Lazy-init ControlSystem (shared) + fresh Simulation per call

9. Remaining Work Before Final Submission

#	Task	Priority	Status
1	Improve model accuracy (deeper architectures, ensembles)	High	Not started
2	Add matplotlib/plotly visualizations to dashboard	Medium	Not started
3	Model comparison (MLP vs Random Forest vs SVM)	Medium	Not started
4	Stricter input validation and error messages	Low	Partial
5	Batch prediction via CSV upload	Low	Not started
6	Polish README and final submission document	High	In progress
7	User testing and usability feedback	Medium	Not started
8	K-fold cross-validation for robust evaluation	Medium	Not started

10. Updated Contribution of Each Member

NOTE: Replace placeholder names with actual team member names and student IDs.

Member Name	Student ID	Contribution	%
Member 1 (Lead)	XXXXXXX	Neural Network module, data cleaning, model training	25%
Member 2	XXXXXXX	Fuzzy Logic engine, membership functions, 36 rules	25%
Member 3	XXXXXXX	Rule-Based engine, recommendations, metric modifiers	25%
Member 4	XXXXXXX	Streamlit dashboard, UI, test pipeline, integration, report	25%

Project File Structure

File	Lines	Size	Role
nn_model.py	115	3.9 KB	Neural network - data cleaning, training, prediction
fuzzy_engine.py	191	9.9 KB	Fuzzy logic - membership functions, inference, 36 rules

File	Lines	Size	Role
rules_engine.py	148	6.9 KB	Rule-based engine - personalized recommendations
app.py	277	12.1 KB	Streamlit dashboard - UI integrating all three modules
test_pipeline.py	219	8.1 KB	End-to-end test suite with 10 test profiles
diabetes.csv	769	24.1 KB	Pima Indians Diabetes Dataset (768 records)
model.pkl	-	25.0 KB	Trained MLPClassifier model (serialized)
scaler.pkl	-	1.1 KB	Fitted StandardScaler (serialized)
requirements.txt	8	132 B	Python dependencies